

MTR-1 Multi-Target Radar Multisensor

Multi-target, multi-zone mmWave presence sensing for Home Assistant

The Apollo MTR-1 is a compact multi-target radar multisensor built for ESPHome and Home Assistant. At its core is an HLK-LD2450 24 GHz mmWave radar that tracks up to three targets across three zones and reports their coordinates — so you know not just whether someone is in a room, but where. It is powered over USB-C, connects over Wi-Fi, and runs entirely on your local network with no cloud or subscription required.

Measuring just 49 x 32 x 15 mm, the MTR-1 packs a diverse sensor array into a low-profile package: light and UV sensing, optional CO₂, onboard RGB and a piezo buzzer for alerts, and Bluetooth tracking. All firmware and CAD files are open source.

Key Specifications

Product	MTR-1 Multi-Target Radar Multisensor (LD2450)
Manufacturer	Apollo Automation (Versailles, KY, USA)
Presence sensor	HLK-LD2450 24 GHz mmWave radar
Targets / zones	Up to 3 targets, 3 zones, with target coordinates
Max range	6 m (19.6 ft)
Microcontroller	ESP32-C3
Firmware	ESPHome — open source
Connectivity	Wi-Fi (2.4 GHz) + Bluetooth
Power input	USB-C, standard 5 V wall adapter
Dimensions	49 x 32 x 15 mm
Onboard feedback	RGB neopixel + piezo buzzer
Local control	Yes — no cloud or subscription required
In the box	Assembled sensor with case; Apollo sticker

Onboard & Optional Sensors

Sensor	Function	Availability
HLK-LD2450	24 GHz mmWave radar — multi-target (3) and multi-zone (3) presence with coordinates	Standard
LTR-390UV	Ambient light (LUX) and ultraviolet (UV) sensing	Standard
SCD40	True CO ₂ sensing — pre-calibrated	Optional
ESP32-C3-Mini	Bluetooth device / beacon tracking (YAML config required)	Standard

The LD2450's wider awareness enables location-based automations — detecting whether someone is at the stove, sink, couch, or in bed — rather than simple room occupancy.

Configurations & Pricing

Configuration	CO ₂ (SCD40)	Price (USD)
Base — mmWave, LUX/UV, RGB, buzzer, Bluetooth	—	\$39.99

Configuration	CO2 (SCD40)	Price (USD)
With CO2	Yes	\$59.99

Features & Use Cases

- **Multi-target presence:** track up to three people at once and trigger lights instantly on entry.
- **Zone & location automations:** define up to three zones to know where a person is — stove, sink, couch, or bed.
- **Light & UV sensing:** use LUX to switch lights only when a room is dark and UV to gauge sunlight.
- **CO₂ awareness (optional):** detect elevated indoor CO₂ affecting sleep and focus; trigger ventilation.
- **Alerts:** RGB pixel and piezo buzzer provide visual and audible notifications and beep patterns.
- **Bluetooth tracking:** locate phones, tags, or beacons by proximity (YAML configuration required).

Important Notes

- **Single-board, multi-target:** the LD2450 tracks up to 3 targets; zones must be squared to the sensor's orientation, and its 120° field of view benefits from planned placement.
- **Exclusion zones:** the sensor is highly sensitive; setting up exclusion zones is recommended to avoid false positives.
- **Rear USB-C adapter:** some wall plugs and receptacles with smart charging may not work with the back male USB-C adapter.
- **Not a safety device:** the optional CO₂ sensor does not meet NBIC, NFPA, or IFC requirements for CO₂ safety monitoring.

Documentation & Resources

- **Product page:** apolloautomation.com/products/mtr-1
- **Wiki / setup guide:** wiki.apolloautomation.com/products/mtr1
- **Source & CAD files:** github.com/ApolloAutomation/MTR-1
- **Community:** [Apollo Automation Discord](https://discord.com/invite/apolloautomation)
- **Sensor datasheets:** Hi-Link LD2450, Liteon LTR-390UV, Sensirion SCD40, Espressif ESP32-C3-Mini.